higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

**NATIONAL CERTIFICATE**

# INDUSTRIAL ELECTRONICS N5

(8080175)

**04 April 2023 (X-paper)**
**09:00–12:00**

**Drawing instruments and nonprogrammable calculators may be used.**

**This question paper consists of 6 pages and a formula sheet of 6 pages.**

112Q1A2304

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# DEPARTMENT OF HIGHER EDUCATION AND TRAINING
# REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
INDUSTRIAL ELECTRONICS N5
TIME: 3 HOURS
MARKS: 100

**INSTRUCTIONS AND INFORMATION**

1. Answer all the questions.
2. Read all the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Start each question on a new page.
5. Use only a black or blue pen.
6. Sketches must be large, neat and fully labelled.
7. Write neatly and legibly.

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## QUESTION 1: ALTERNATING CURRENT THEORY

1.1 Draw a neat graphical representation to illustrate the following:

1.1.1 A time constant in which a change equal to 63.2% will take place

1.1.2 Input and output waveforms of a coupling circuit

(2 × 2) (4)

1.2 Explain the purpose of a capacitor coupling circuit. (2)

1.3 Explain in your own words why square waves are suitable for testing certain electronic circuits. (2)

1.4 A series resonance network consists of the following components:

Resistor: $R_1 = 15\ \Omega$

Inductor: $L_1 = 40$ mH

Voltage: $V_T = 20$ V

Frequency: f = 1kHz

Calculate the value of the capacitor needed at resonance. (3)

**[11]**

## QUESTION 2: POWER SUPPLIES

2.1 A voltage reference circuit of an instrumentation control system has the following specifications:

Zener diode uses 10 V, 600 mW

Supply voltage ($V_S$) = 150 V +/- 1 V DC

Calculate:

2.1.1 Minimum value of series resistor (4)

2.1.2 Minimum value of load resistor (2)

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2.2 FIGURES 1(a) and (b) below show the two versions of the integrated 7812 regulators commonly used. Label each regulator by writing only the answer next to the number (1–3) in the ANSWER BOOK:

2.2.1

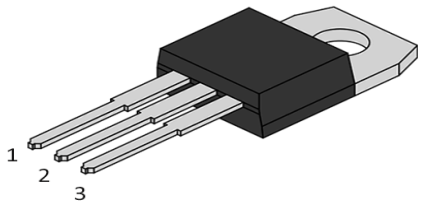



**FIGURE 1(a) TO-220 7812 regulator**

(3 × 1)

2.2.2

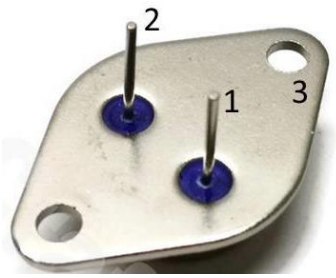



**FIGURE 1(b) TO-3 7812 regulator**

(3 × 1) (6)

2.3 Draw a neat, labelled circuit diagram of a basic series control circuit. (4)

2.4 Name **TWO** types of regulators used in power supplies. (2)
**[18]**

## QUESTION 3: TRANSISTOR AMPLIFIERS

3.1 A common emitter amplifier used in a radio frequency circuit has the following specifications:

+Vcc = 12V; $I_C$ = 10mA; Vce = 6V;
$\beta$ = 250 Re = 120$\Omega$ and Vbe = 0,6V

Calculate the values of the following components:

3.1.1 RC (resistor-capacitor) (4)

3.1.2 $R_{b1}$ and $R_{b2}$ (6)

3.2 Calculate the input impedance $Z_i$ and the output impedance $Z_O$ of the radio frequency circuit in QUESTION 3.1 by means of the approximate method if:

$h_{ie}$ = 1,2kΩ; $h_{re} = 2 \times 10^{-4}$;
$h_{fe}$ = 100; and $h_{oe}$ = 20μA/V ($R_S$ = 0) (4)

3.3 What effect does a temperature rise have upon the following?

3.3.1 The collector current $I_{CO}$

3.3.2 The base-emitter voltage $V_{be}$

(2 × 2) (4)
**[18]**

## QUESTION 4: OPERATIONAL AMPLIFIERS

4.1 Draw a neat, labelled circuit diagram of a practical operational amplifier connected as a differentiator. (4)

4.2 Calculate the value of the capacitor in QUESTION 4.1 if $R_2$ = 2 kΩ and the work frequency (f) = 500 Hz. (4)
**[8]**

## QUESTION 5: INTEGRATED CIRCUITS

5.1 What should be done with the unused inputs of a CMOS-integrated circuit when it is connected to a circuit? (2)

5.2 Indicate whether the following statements are TRUE or FALSE by writing only 'True' or 'False' next to the question number (5.2.1–5.2.3) in the ANSWER BOOK.

5.2.1 CMOS-integrated circuits have higher noise immunity.

5.2.2 CMOS-integrated circuits are susceptible to static charges because of their low reactive input.

5.2.3 As soon as work is done on the circuit of CMOS-integrated circuits, the power supply to the circuit must be switched off.

(3 × 1) (3)
**[5]**

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## QUESTION 6: ELECTRONIC PHASE CONTROL

6.1 Draw a neat, labelled diagram of the Hall effect displacement transducer. (6)

6.2 Briefly explain the operating principle of a transmitter/receiver system that uses an infrared diode and a photodiode. (6)
**[12]**

## QUESTION 7: TEST EQUIPMENT

Explain, with the aid of a sketch of a waveform, what is meant by *a half-controlled bridge rectifier circuit*. **[5]**

## QUESTION 8: OSCILLATORS

8.1 Explain the difference between the tuned (tank) circuit of the Hartley oscillator and Collpits oscillator. (4)

8.2 Describe the feedback phase displacement in the RC network of an RC phase displacement oscillator. (6)
**[10]**

## QUESTION 9: TRANSDUCERS

9.1 Draw the circuit diagram of a thermistor control circuit that makes use of an operational amplifier and a DC Wheatstone bridge. (4)

9.2 The bridge is balanced at 25 °C; A = 0,2169; $\beta$ = 3 200 and a 12 V battery is connected across the bridge; $R_t$ = 10 k$\Omega$ at 25 °C.

Calculate the following:

9.2.1 The value of the thermistor at 30 °C (2)

9.2.2 The gain of the amplifier with an output of 10 V (7)
**[13]**

**TOTAL: 100**

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## FORMULA SHEET

$$I = \frac{V}{R} \qquad P = IV = I^2R = \frac{V^2}{R}$$

$$V_T = V_1 + V_2 + V_3 + ... = I_1R_1 + I_2R_2 + I_3R_3 + ...$$

$$I_T = I_1 + I_2 + I_3 + ... = \frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} + ...$$

$$T = RC \qquad T = \frac{L}{R}$$

$$V_R = RC\frac{dv}{dt} \qquad V_C = \frac{1}{RC}\int v_i dt$$

$$X_L = 2\pi fL \qquad X_C = \frac{1}{2\pi fC}$$

$$Z = R + jX_L \qquad Z = R - jX_C$$

$$Z = R + j(X_L - X_C) \qquad I_T = \frac{V_T}{Z_T}$$

$$V_R = I_TR \qquad V_L = I_T(jX_L)$$

$$V_C = I_T(-jX_C) \qquad f_r = \frac{1}{2\pi\sqrt{LC}}$$

$$Q = \frac{V_L}{V_T} = \frac{V_C}{V_T} = \frac{X_L}{R} = \frac{X_C}{R} = \frac{1}{R}\sqrt{\frac{L}{C}} = \frac{f_r}{f_2 - f_1}$$

$$BW = f_2 - f_1 \qquad \frac{1}{Z_T} = \frac{1}{Z_1} + \frac{1}{Z_2}$$

$$Z_T = \frac{Z_2Z_2}{Z_1 + Z_2} \qquad I_T = I_1 + I_2 = \frac{V}{Z_1} + \frac{V}{Z_2}$$

$$Z_T = \frac{R(jX_L)}{R + jX_L} \qquad \frac{1}{Z_T} = \frac{1}{R} - \frac{j}{X_L}$$

$$I_T = I_R - jI_L \qquad I_T = \frac{V}{R} - j\frac{V}{X_L}$$

$$Z_T = \frac{R(-jX_C)}{R - jX_C} \qquad \frac{1}{Z_T} = \frac{1}{R} + \frac{j}{X_C}$$

$$I_T = I_R + jI_C \qquad I_T = \frac{V}{R} + j\frac{V}{X_C}$$

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$$\frac{1}{Z_T} = \frac{1}{R} - j\left(\frac{1}{X_L} - \frac{1}{X_C}\right)$$

$$I_T = I_R - j(I_L - I_C) \qquad I_T = \frac{V}{R} - j\left(\frac{V}{X_L} - \frac{V}{X_C}\right)$$

$$a + jb = \sqrt{a^2 + b^2} \,/\, \underline{\tan^{-1}\frac{b}{a}} = r \,/\underline{\theta}$$

$$r/\underline{\theta} = r(\cos\theta + j\sin\theta) \qquad Q = \tan\theta$$

$$f = \frac{1}{2\pi}\sqrt{\frac{1}{LC} - \frac{R^2}{L^2}} \qquad Z_d = \frac{L}{CR_1}$$

$$V_{rms} = \frac{1}{\sqrt{2}} V_m = 0{,}707\, V_m \qquad V_{dc} = \frac{2}{\pi} V_m = 0{,}637\, V_m$$

$$\frac{V_P}{V_S} = \frac{N_P}{N_S} = \frac{I_S}{I_P} \qquad V_{dc} = \frac{1}{\pi} V_m = 0{,}318\, V_m$$

$$PIV = V_m \qquad PIV = 2\, V_m$$

$$R_{r(rms)} = 0{,}385\, V_m \qquad V_{r(rms)} = 0{,}305\, V_m$$

$$r = \frac{V_{r(rms)}}{V_{dc}} \qquad V_{r(rms)} = \frac{V_{r(p-p)}}{2\sqrt{3}}$$

$$V_{dc} = V_m - \frac{V_{r(p-p)}}{2}$$

$$V_{dc} = V_m - \frac{I_{dc}}{2fC} \qquad V_{dc} = V_m - \frac{I_{dc}}{4fC}$$

$$V_{r(rms)} = \frac{I_{dc}}{2\sqrt{3}fC} = \frac{V_{dc}}{2\sqrt{3}fCR_L} \qquad V_{r(rms)} = \frac{I_{dc}}{4\sqrt{3}fC} = \frac{V_{dc}}{4\sqrt{3}fCR_L}$$

$$r = \frac{I_{dc}}{2\sqrt{3}fCV_{dc}} = \frac{1}{2\sqrt{3}fCR_L} \qquad r = \frac{I_{dc}}{4\sqrt{3}fCV_{dc}} = \frac{1}{4\sqrt{3}fCR_L}$$

$$V'_{dc} = \frac{R_L}{R + R_L} . V_{dc} \qquad V'_{r(rms)} = \frac{X_C}{\sqrt{R^2 + X_C^2}} \cdot V_{r(rms)}$$

$$X_C = \frac{1}{2\pi fC} \qquad X_C = \frac{1}{4\pi fC} \qquad r' = \frac{V'_{r(rms)}}{V'_{dc}}$$

$$V'_{r(rms)} = \frac{X_C}{R} \cdot V_{r(rms)} \qquad r' = rX_C\left(\frac{R + R_L}{R.R_L}\right)$$

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$$TV'_{dc} = V_{dc} - I_{dc}R_1$$

$$V'_{dc} = \frac{R_L}{R_L + R_1} . V_{dc}$$

$$V'_{r(rms)} = \frac{V_{r(rms)}}{(2\pi f)^2 LC}$$

$$V'_{r(rms)} = \frac{V_{r(rm)}}{(4\pi f)^2 LC}$$

$$VR = \frac{V_{NL} - V_{FL}}{V_{FL}}$$

$$\%VR = \frac{V_{NL} - V_{FL}}{V_{FL}} \times 100$$

$$2V_m = V_{c2} = V_m + V_{c1}$$

$$3V_m = V_{c1} + V_{c3} = V_m + 2V_m$$

$$S = \frac{\Delta V_o}{\Delta V_i}$$

$$V_R = V_i - V_z$$

$$R_{s(\min)} = \frac{V_{i(\max)} - V_z}{I_{z(\max)}}$$

$$I_z = \frac{P_z}{V_z}$$

$$R_{L(\min)} = \frac{V_Z}{V_{i(\max)} - V_Z} . R_S$$

$$V_o = V_r - V_{be}$$

$$R_c = \frac{V_{cc} - V_{ce}}{I_c}$$

$$R_b = \frac{V_{cc} - V_{be}}{I_b}$$

$$\beta = \frac{I_c}{I_b}$$

$$C_e \geq \frac{10}{2\pi f R_e}$$

$$V_e = \frac{V_{cc}}{10}$$

$$R_e = \frac{V_e}{I_e} \simeq \frac{V_e}{I_c}$$

$$R_c = \frac{V_{cc} - V_{ce} - V_e}{I_c}$$

$$R_b = \frac{V_{cc} - V_{be} - V_e}{I_b}$$

$$R_{b1} = \frac{R_{b2}(V_{cc} - V_b)}{V_b}$$

$$R_{b2} = \frac{1}{10}\beta R_e$$

$$V_b = V_e + V_{be}$$

$$V_{be} = h_{ie} i_b + h_{re} V_{ce}$$

$$i_c = i_{fe} i_b + h_{oe} V_{ce}$$

$$A_i = \frac{h_{fe}}{1 + h_{oe} Z_L}$$

$$A_i = h_{fe}$$

$$A_i = \left(\frac{h_{fe}}{1 + h_{oe} Z_L}\right)\left(\frac{R_b T}{R_{bT} + Z_1}\right)\left(\frac{R_c}{R_c + R_L}\right)$$

$$A_v = \frac{-h_{fe} Z_L}{h_{ie} + (h_{ie} h_{oe} - h_{fe} h_{re}) Z_L}$$

$$A_v = \frac{-h_{fe} Z_L}{h_{ie}}$$

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$$Z_1 = h_{ie} - \frac{h_{fe}h_{re}Z_L}{1 + h_{oe}Z_L} \qquad Z_1 = h_{ie}$$

$$Z_2 = \frac{1}{h_{oe} - \frac{h_{fe}h_{re}}{h_{ie} + R_s}} \qquad Z_2 = \frac{1}{h_{oe}}$$

$$A_p = \frac{A_i^2 R_L}{R_1} = -A_v A_i \qquad A_p = \frac{{h_{fe}}^2 R_L}{h_{ie}}$$

$$Z_0 = R_C // R_L // Z_2 = Z_L // Z_2$$

$$Z_0 = R_C // Z_2 = Z_L // Z_2 \qquad Z_1 = R_b // Z_1$$

$$Z_i = R_{b1} // R_{b2} // Z_1 \qquad I_1 = \frac{R_{bT} I_i}{R_{bT} = Z_1}$$

$$I_0 = h_{fe} I_b = h_{fe}\left(\frac{R_{b2}(I_i)}{R_{b2} + h_{ie}}\right) \qquad A_i = \frac{I_0}{I_1}$$

*For common base, substitute all the 'e' subscripts with a 'b' in the h-parameters.*

$$Z_L = R_c // R_L \qquad I_1 = \frac{R_e I_i}{R_e + Z_1}$$

$$CMRR = \frac{A_{dm}}{A_{cm}} \qquad CMRR\ (dB) = 20log\frac{A_{dm}}{A_{cm}}$$

$$I_e = \frac{V_e}{R_e} \qquad I_c = \frac{I_e}{2}$$

$$R_L = \frac{V_{R_L}}{I_C} \qquad g_m R_L = \frac{h_{fe}}{h_{ie}} . R_L$$

$$V_0 = -\left(\frac{R_f}{R_1}\right) . V_i \qquad V_0 = \left(\frac{R_f}{R_1} + 1\right) . V_i$$

$$V_0 = -\left(\frac{R_f}{R_1} . V_1 + \frac{R_f}{R_2} . V_2 + \frac{R_f}{R_3} . V_3\right) \quad V_0 = -\left(\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3}\right) R_f$$

$$V_0 = -(V_1 + V_2 + V_3) \qquad V_0 = -(I_1 + I_2 + I_3)\ R_f$$

$$V_0(t) = -\frac{1}{RC}\int V_i(t) \qquad V_0(t_b) = -\frac{1}{RC}\int_{t_a}^{t_b} V_i(t_b) + V_c(t_a)$$

$$t = \frac{1}{f} \qquad A_v = -\frac{R_s}{R_1}$$

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$$R_2 = \frac{R_1 R_s}{R_1 + R_s}$$

$$f_c = \frac{1}{2\pi R_s C}$$

$$V_0(t) = -RC\frac{dV_i(t)}{dt}$$

$$A = \frac{R_f}{X_c}$$

$$A = -\frac{R_f}{R_s}$$

$$V_0(t) = -R_f C\frac{d}{dt}.v_i \sin \omega t$$

$$t = R_f C$$

$$V_0 = A(V_r - V_i)$$

$$V_0 = \frac{R_f}{R_s}(V_2 - V_1)$$

$$V_0 = V_2 - V_1$$

$$f_0 = \frac{1}{2\pi\sqrt{C_1 C_2 R_1 R_2}}$$

$$f_0 = \frac{1}{2\pi RC}$$

$$f_0 = \frac{1}{2\pi\sqrt{L_T C_1}}$$

$$L_T = L_1 + L_2 + 2M$$

$$f_0 = \frac{1}{2\pi\sqrt{LC_T}}$$

$$C_T = \frac{C_1 C_2}{C_1 + C_2}$$

$$f_0 = \frac{1}{2\pi\sqrt{LC_2}}$$

$$f = \frac{1}{2\pi RC\sqrt{6}}$$

$$f_0 = \frac{1{,}5}{RC}$$

$$f_0 = \frac{1}{t} = \frac{1}{t_1 + t_2}$$

$$t_1 = 0{,}7\ R_2 C_1$$

$$t_2 = 0{,}7\ R_1 C_2$$

$$f_0 = \frac{1}{1{,}4RC}$$

$$V_i = I_{c2} R_e + V_{be(ON)}$$

$$t = 1{,}1\ RC$$

$$f_0 = \frac{1{,}443}{(R_A + 2R_B)C}$$

$$t_{low} = 0{,}693\,(R_B)C$$

$$t_{high} = 0{,}693\,(R_A + R_B)C$$

$$t_T = t_{low} + t_{high}$$

$$K = \frac{\Delta R / R}{\Delta l / l}$$

$$\sigma = \Delta l / l$$

$$R = \rho\frac{1}{\pi d^2 / 4}$$

$$\sigma = \frac{S}{E}$$

$$Resolution = \frac{1}{amount\ of\ turns}$$

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$$Resolution = \frac{voltage\ drop\ across\ adjacent\ turns}{total\ voltage\ drop}$$

$$R_t = Ae^{B/T} \qquad T = 273 + °C$$

$$V_A = \frac{R_2}{R_1 + R_2} . V_T \qquad V_B = \frac{R_t}{R_t + R_3} . V_T$$

$$V_{AB} = V_A - V_B \qquad A_v = \frac{V_0}{V_i}$$

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